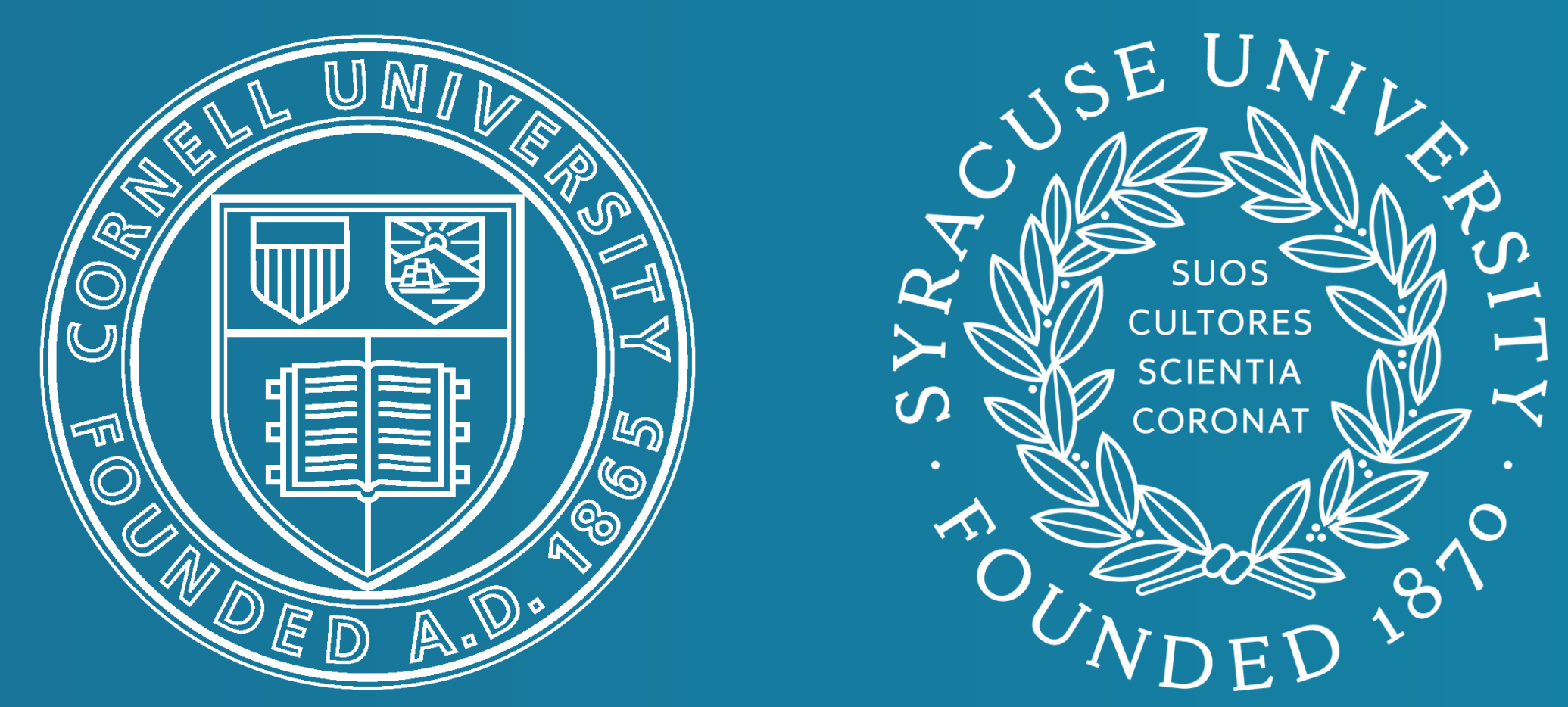


Knowledge Graphs for Solar Energy News Analysis

Rachel Kan, Chris Dunham
Cornell University, Syracuse University



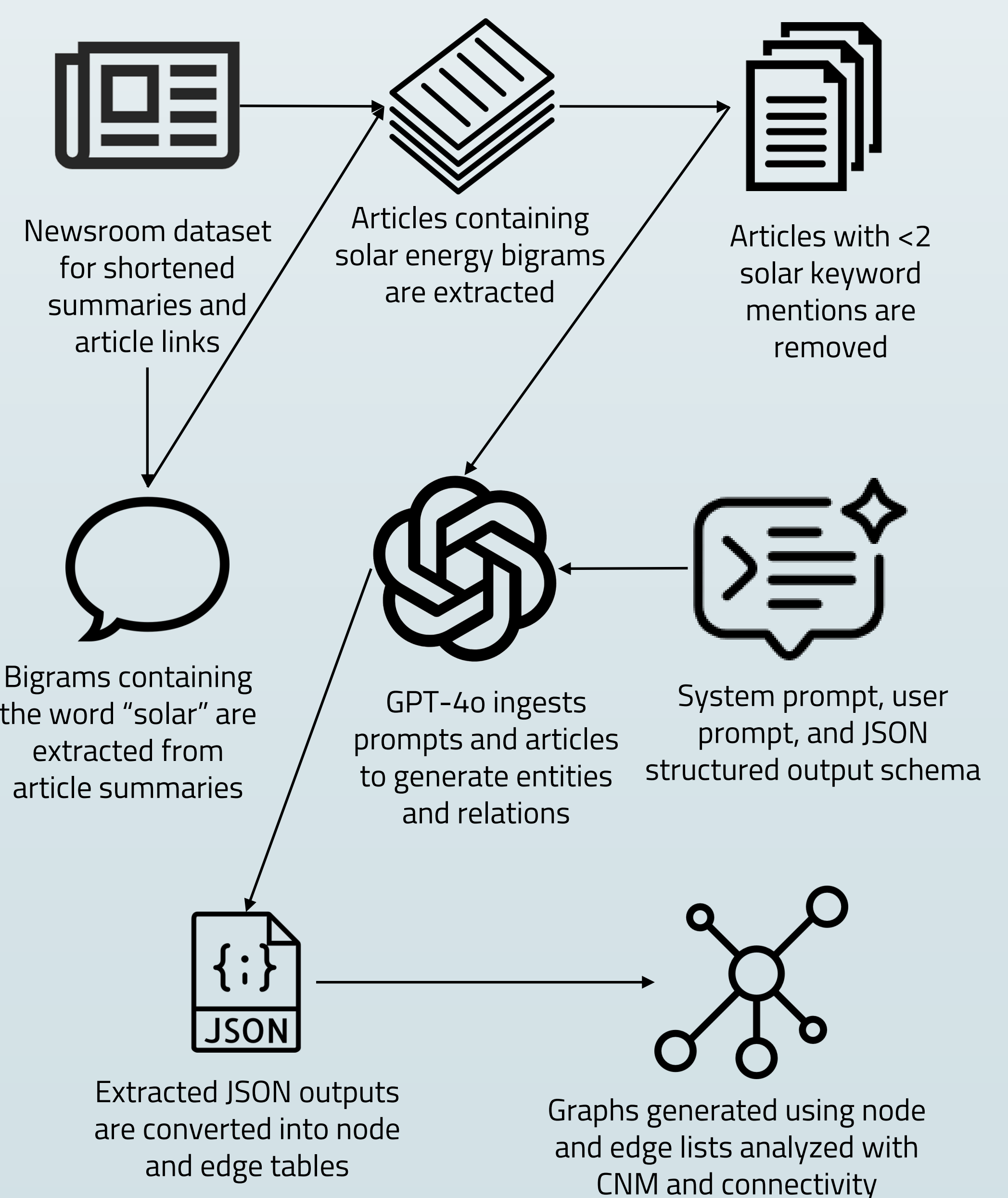
Introduction

The global solar energy industry consists of interconnected organizations, technologies, policies, and markets that are difficult to analyze from unstructured information sources. This project uses OpenAI’s GPT-4o to extract entities and relationships from Cornell Newsroom articles [1] and construct a solar energy knowledge graph. Graph analysis and community detection are then applied to identify major themes, influential entities, and connections within the international solar energy ecosystem.

Questions

- How do solar energy industries interact across world borders? In particular, what types of entities connect these communities directly and indirectly?
- How can information be extracted from a large corpus of articles to analyze communities and relationships in the solar energy industry?

Methodology



Methodology

Data Collection/Filtering
Cornell Newsroom articles were screened using domain-specific solar energy keywords and scraped for full-text analysis. Articles with fewer than two solar keyword mentions were excluded, reducing GPT-4o token costs (Figure 1).

Item	Before	After
Number of articles	5,374	1,985
System prompt tokens	1,773	775
Article corpus tokens	6,673,300	2,534,844
Total input tokens	16,201,402	4,073,219
Estimated input cost	\$40.51	\$10.19

Fig. 1: Reductions in dataset size and cost with filtering pipeline

LLM-Based Information Extraction
GPT-4o was used to extract entities and semantic relationships from solar energy articles, as LLMs outperform traditional methods on network extraction tasks [2]. A structured prompting framework (system prompt, user prompt, and JSON schema) defined 8 node types and 19 edge types while optimizing token usage.

Knowledge Graph Construction
Extracted JSON outputs were converted into a knowledge graph with 4,002 nodes and 5,127 edges. Low-degree nodes were removed, similar entities merged with fuzzy matching and transformer-based similarity, and node size scaled by degree.

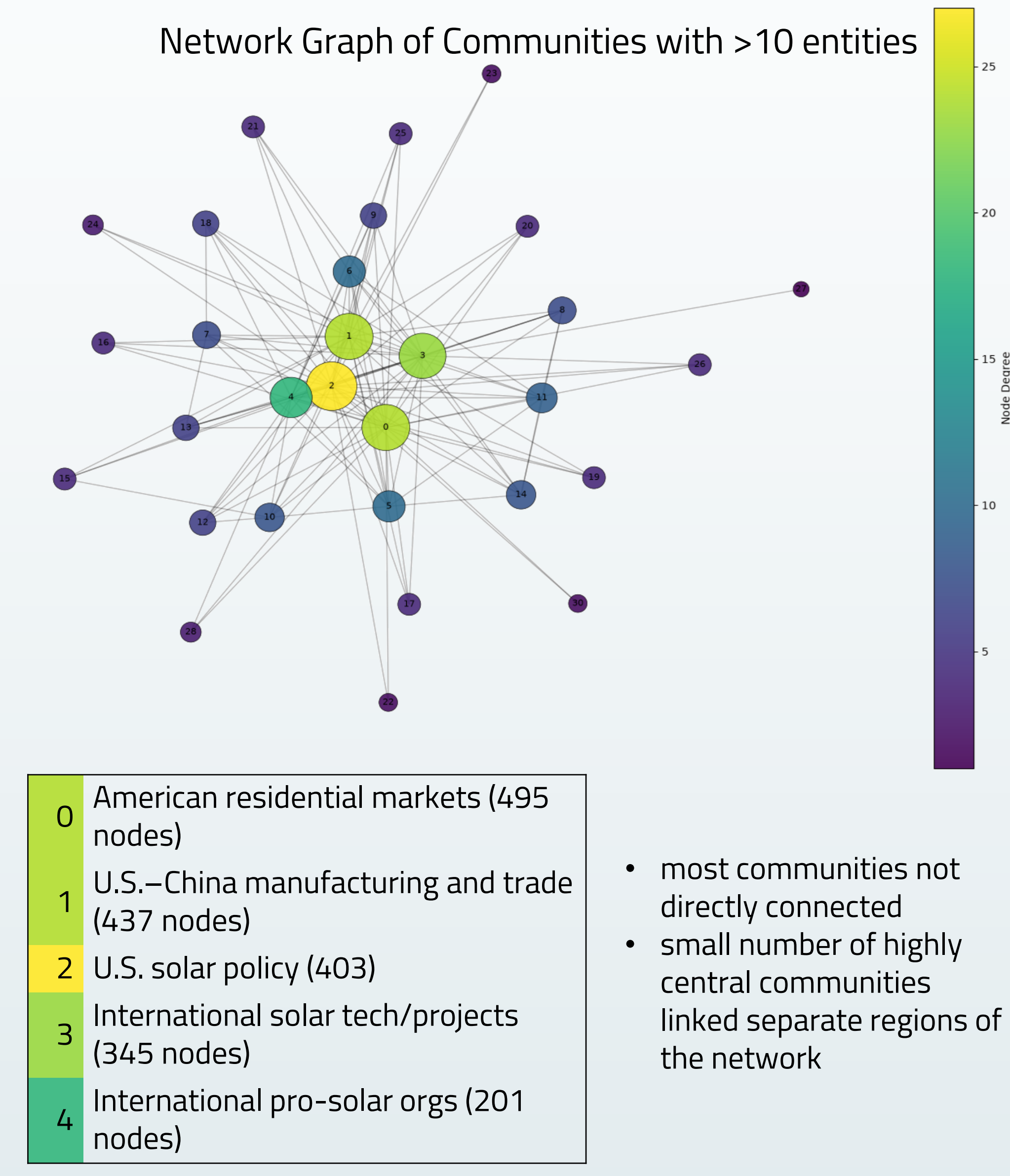
Network Analysis
The greedy modularity maximization/Clauset-Newman-Moore (CNM) community detection algorithm was applied to identify groups of closely connected entities. 408 communities were found, of which 31 had over 10 member entities.

Source Type	Target Type	Count
Company	Company	79
Technology	Company	66
Company	Project	57
Company	Technology	46
Company	Location	40
Technology	Organization	35
Location	Project	29
Technology	Location	25
Company	Policy	24

Company, technology, and project nodes appear most frequently in common edges, while government, policy, people, and article nodes are less common, suggesting news coverage emphasizes technological and economic developments over politics.

Fig. 2: Counts of most common edge connection types

Results



CNM revealed that clusters formed primarily around geographic regions and areas of focus (e.g. technology development, policy, manufacturing, and international collaborations).

Node Type	Node	Inter-Community Links
Technology	solar_panels	100
Technology	solar_energy	54
Company	first_solar	50
Location	california	32
Location	united_states	28
Company	tesla_motors	27
GovernmentAgency	department_of_energy	26
Company	sunpower_corporation	24
Company	solarcity	22
Location	china	21

Fig. 3: Nodes with highest connectivity to other communities

Though company, technology, and project nodes are most often found in common edges, location nodes play a large role in the connections between communities.

Limitations

- Articles are limited to 1998–2017, preventing analysis of recent trends (e.g., artificial intelligence).
- The dataset also includes only 38 major publications and excludes non-English sources, limiting coverage and diversity.

Future Work

- Future work includes analyzing reporting patterns across news organizations (e.g., topics, geographic focus, and coverage).
- The same pipeline can also be applied to other topics using topic-specific filtering and network analysis.
- The solar energy knowledge graph may be used to ground LLM outputs in a compound AI system [3].

References

[1] M. Grusky, M. Naaman, and Y. Artzi, “NEWSROOM: A Dataset of 1.3 Million Summaries with Diverse Extractive Strategies,” 2018. Accessed: Aug. 04, 2026. [Online]. Available: <https://aclanthology.org/N18-1065.pdf>

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[3] T. Ma, B. Liu, and S. Yang, “Constructing New Power System Stock Indices via a Compound AI System: A Hierarchical RAG and Knowledge Graph Approach,” *Proceedings of the 2025 6th International Conference on Computer Science and Management Technology*, pp. 1007–1012, Dec. 2025, doi: 10.1145/3795154.3795318.

For more technical details, please see the project’s Github: <https://github.com/rckan/syracuse-su26-project>

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